

Equivalence Between Principal Component Analysis (PCA) and the Moment of Inertia Tensor in Classical Mechanics

1 Similarity

PCA and the moment of inertia tensor describe the same mathematical and geometric idea in two different fields.

Both problems solve an **eigenvalue and eigenvector problem** by diagonalizing a real, symmetric, and positive semi-definite matrix.

- **Principal Component Analysis (PCA):** The main goal of PCA is to find perpendicular directions (principal components) that maximize the variance in a multidimensional dataset. The data cloud forms an ellipsoid in space, and PCA finds the main axes of this ellipsoid.
- **Moment of Inertia Tensor:** In classical mechanics, the inertia tensor describes how a rigid body resists rotation around a given axis based on its mass distribution. The body also forms a mass distribution ellipsoid in space.

Physically, these two concepts are exact complements of each other. If a mass or data distribution spreads out further along a specific axis (maximum variance and largest eigenvalue in PCA), its moment of inertia around that axis is at its minimum.

In other words, the direction where the data spreads the most is the easiest axis for the body to rotate around.

Both problems come from optimizing the form $A\mathbf{v} = \lambda\mathbf{v}$.

Table 1: PCA vs. Inertia Tensor Terminology

PCA	Inertia Tensor
Scatter / Covariance Matrix (S)	Moment of Inertia Tensor (I)
Variances (x^2, y^2)	Moments of Inertia ($y^2 + z^2, x^2 + z^2$)
Covariances (xy)	Products of Inertia ($-xy$)
Principal Components ($\mathbf{w}_i$)	Principal Axes ($\boldsymbol{\omega}_i$)
Eigenvalues / Variances (λ_i)	Principal Moments of Inertia (γ_i)
Maximum Variance Direction ($\max \lambda$)	Minimum Rotational Axis ($\min \gamma$)

2 PCA

Let $X \in \mathbb{R}^{N \times d}$ be a zero-mean (centered) data matrix with N observations and d variables. The covariance matrix Σ , which shows how variables change together, is defined as:

$$\Sigma = \frac{1}{N-1} X^T X \quad (1)$$

PCA looks for a unit vector $\mathbf{w}$ that maximizes the variance of the projected data:

$$\max_{\mathbf{w}} \mathbf{w}^T \Sigma \mathbf{w} \quad \text{subject to} \quad \mathbf{w}^T \mathbf{w} = 1 \quad (2)$$

We solve this constrained optimization problem using Lagrange multipliers:

$$\mathcal{L}(\mathbf{w}, \lambda) = \mathbf{w}^T \Sigma \mathbf{w} - \lambda(\mathbf{w}^T \mathbf{w} - 1) \quad (3)$$

Taking the derivative with respect to $\mathbf{w}$ and setting it to zero gives the eigenvalue equation:

$$\Sigma \mathbf{w} = \lambda \mathbf{w} \quad (4)$$

Here, the vectors $\mathbf{w}$ are the principal component directions (eigenvectors), and λ values are the variances along these directions (eigenvalues).

3 Moment of Inertia and Principal Axes Calculation

For a rigid body or a discrete system of N point masses with positions $\mathbf{r}_k = [x_k, y_k, z_k]^T$ and masses m_k , the inertia tensor $I \in \mathbb{R}^{3 \times 3}$ is defined in tensor form as:

$$I = \sum_{k=1}^N m_k ((\mathbf{r}_k^T \mathbf{r}_k) \mathbb{I}_{3 \times 3} - \mathbf{r}_k \mathbf{r}_k^T) \quad (5)$$

In matrix component form, it is written as:

$$I = \begin{pmatrix} \sum m_k (y_k^2 + z_k^2) & -\sum m_k x_k y_k & -\sum m_k x_k z_k \\ -\sum m_k x_k y_k & \sum m_k (x_k^2 + z_k^2) & -\sum m_k y_k z_k \\ -\sum m_k x_k z_k & -\sum m_k y_k z_k & \sum m_k (x_k^2 + y_k^2) \end{pmatrix} \quad (6)$$

The axes where the angular momentum is parallel to the rotation axis are called the "principal axes." Finding these axes means diagonalizing the inertia tensor:

$$I \boldsymbol{\omega} = \gamma \boldsymbol{\omega} \quad (7)$$

Here, ω represents the principal axes directions, and γ represents the principal moments of inertia.

If we define the scatter matrix $S = \sum m_k \mathbf{r}_k \mathbf{r}_k^T$ (which is the core of the covariance matrix), we can directly see the relationship between the inertia tensor and PCA:

$$I = \text{Tr}(S) \mathbb{I}_{3 \times 3} - S \quad (8)$$

4 Example

Consider a 2D planar ($z = 0$) mass/data distribution with 4 points centered at zero: $\mathbf{p}_1 = (1, 2, 0)$, $\mathbf{p}_2 = (-1, -2, 0)$, $\mathbf{p}_3 = (2, 1, 0)$, and $\mathbf{p}_4 = (-2, -1, 0)$. We assume $m_k = 1$ for all points.

Step 1: Constructing the Covariance (Scatter) and Inertia Matrices

Sums of squares and cross-products:

$$\begin{aligned} \sum x^2 &= 1 + 1 + 4 + 4 = 10 \\ \sum y^2 &= 4 + 4 + 1 + 1 = 10 \\ \sum xy &= 2 + 2 + 2 + 2 = 8 \end{aligned}$$

Scatter matrix (core PCA matrix S):

$$S = \begin{pmatrix} 10 & 8 & 0 \\ 8 & 10 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (9)$$

Moment of inertia tensor I :

$$\begin{aligned} I_{xx} &= \sum y^2 = 10 \\ I_{yy} &= \sum x^2 = 10 \\ I_{zz} &= \sum (x^2 + y^2) = 20 \\ I_{xy} &= -\sum xy = -8 \end{aligned}$$

$$I = \begin{pmatrix} 10 & -8 & 0 \\ -8 & 10 & 0 \\ 0 & 0 & 20 \end{pmatrix} \quad (10)$$

Step 2: Solving the Eigenvalue Equation

Characteristic equation for PCA (S), $\det(S - \lambda \mathbb{I}) = 0$:

$$(10 - \lambda)^2 - 64 = 0 \implies \lambda^2 - 20\lambda + 36 = 0 \implies (\lambda - 18)(\lambda - 2) = 0 \quad (11)$$

Eigenvalues: $\lambda_1 = 18, \lambda_2 = 2, \lambda_3 = 0$.

Characteristic equation for the inertia tensor (I), $\det(I - \gamma \mathbb{I}) = 0$:

$$(10 - \gamma)^2 - 64 = 0 \quad (\text{with } \gamma_3 = 20 \text{ independently}) \quad (12)$$

Eigenvalues: $\gamma_1 = 2, \gamma_2 = 18, \gamma_3 = 20$.

Note that the relation $\gamma_i = 20 - \lambda_i$ holds exactly, matching $I = \text{Tr}(S)\mathbb{I} - S$.

Step 3: Finding Principal Component Directions and Principal Axes

For the direction of maximum variance ($\lambda_1 = 18$) using $S\mathbf{w}_1 = 18\mathbf{w}_1$:

$$\begin{pmatrix} -8 & 8 & 0 \\ 8 & -8 & 0 \\ 0 & 0 & -18 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \implies x = y, z = 0 \quad (13)$$

Unit vector (First Principal Component): $\mathbf{w}_1 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right)^T$.

For the direction of minimum moment of inertia ($\gamma_1 = 2$) using $I\omega_1 = 2\omega_1$:

$$\begin{pmatrix} 8 & -8 & 0 \\ -8 & 8 & 0 \\ 0 & 0 & 18 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \implies x = y, z = 0 \quad (14)$$

Unit vector (First Principal Axis): $\omega_1 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right)^T$.

Step 4: Comparative Analysis

The results prove the mathematical equivalence:

1. The first principal component ($\mathbf{w}_1$) found by PCA is the direction along which the data cloud spreads the most (maximum variance $\lambda_1 = 18$).
2. The first principal axis (ω_1) found by the inertia tensor is the exact same direction, but it corresponds to the minimum principal moment of inertia ($\gamma_1 = 2$).
3. Mathematically, variance uses inner products of position coordinates ($+x^2$), while inertia uses perpendicular distances ($-x^2$). Because of this, the largest eigenvalue of S creates the smallest eigenvalue of I . Both methods map symmetric matrices to the same orthogonal base (eigenvector space).